

Google Tag Manager (GTM) Event Schema & GA4 Tracking Strategy

OrthoNow Website
Developer Assignment – Task 1

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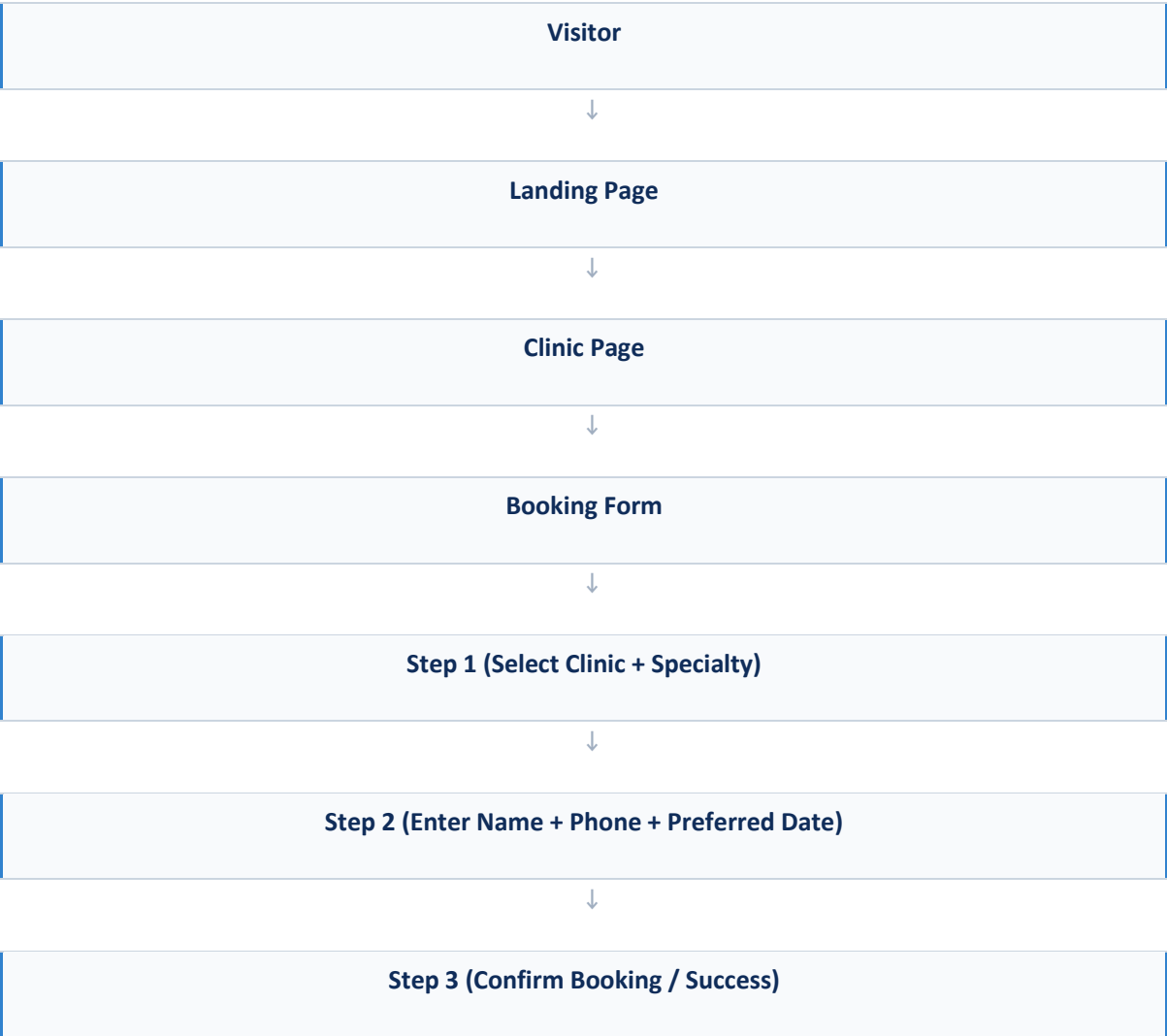
1. Introduction

Before launching paid marketing campaigns, OrthoNow needs reliable tracking to understand how users interact with the website. This document defines a Google Tag Manager (GTM) event tracking strategy for key user actions such as appointment bookings, phone calls, WhatsApp interactions, patient guide downloads, clinic page visits, and blog engagement.

The proposed event schema is designed for Google Analytics 4 (GA4) and Google Ads. It focuses on tracking meaningful business actions that help the marketing team measure campaign performance, identify booking funnel drop-offs, and optimize conversions.

2. Website User Journey Architecture

The diagram below maps the sequence of entry points and actions a visitor travels through across the web application, leading directly to the interactive multi-step form framework:



3. Complete GTM Event Schema Matrix

The primary appointment booking events defined below are custom GA4 events created specifically for OrthoNow's multi-step architecture to capture detailed funnel milestones:

Event Name	Trigger Type	Parameters Captured	GA4 Report Location	Target Audience Group	Operational Note
appointment_booking_start	Custom Event	form_id, booking_session_id	Engagement	All Website Visitors	Fires when user interacts with the first field in Step 1.
appointment_booking_step	Custom Event	form_id, booking_session_id, step_number, step_name, clinic_location, specialty	Engagement	Funnel Progressing Group	Tracks progress through Step 1 and Step 2.
appointment_booking_complete	Custom Event	form_id, booking_session_id, clinic_location, specialty, appointment_id	Conversions	Completed Booking Segment	Final successful booking completion. Used as the primary bidding conversion.
click_to_call	Just Links	click_url, link_text, page_location, clinic_location	Conversions (Secondary)	Phone Outreach Intenders	Tracks direct phone call click interactions across pages.
whatsapp_click	Just Links	click_url, link_text, page_location	Engagement	Chat Inbound Users	Tracks widget engagement launching wa.me interactions.
patient_guide_lead	Custom Event	form_id, guide_name, page_location	Conversions (Secondary)	Information Document Enquirers	Tracks lead capture form completions for downloads.
file_download	Link Click	file_extension, file_name, link_url	Engagement	Content Retainers	Fires when the user downloads the PDF post form submit.

Event Name	Trigger Type	Parameters Captured	GA4 Report Location	Target Audience Group	Operational Note
clinic_page_view	Page View	page_location, page_title, clinic_id, clinic_location	Engagement	Local Regional Intenders	Measures unique traffic patterns to clinic layout pages.
blog_scroll	Scroll Depth	page_location, scroll_threshold, article_title	Engagement	Active Content Consumers	Tracks structural read depths at 25%, 50%, and 75% marks via native GTM scroll trigger.
blog_read_complete	Custom Event	page_location, article_title, time_on_page	Engagement	Deep Content Advocates	Fires exactly when a user reaches a 90% scroll threshold AND spends a minimum of 45 seconds on the article page.

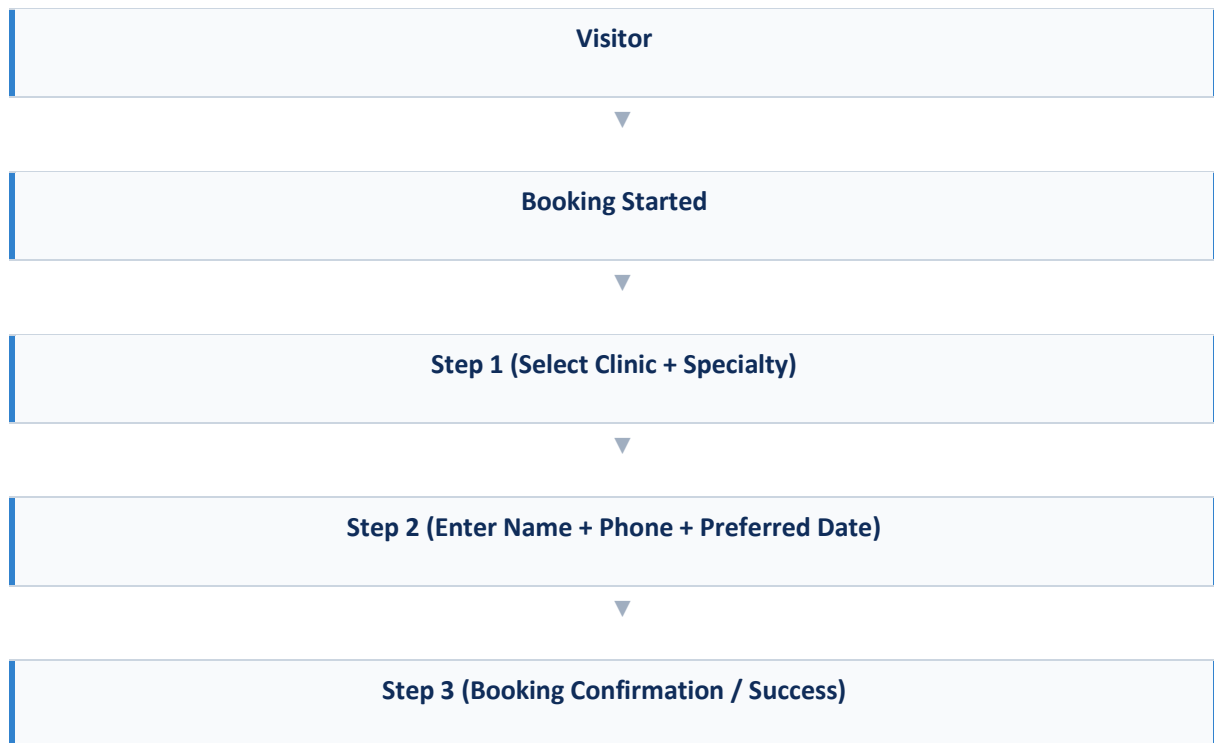
4. Business Value of Event Tracking Strategy

The selected events focus on business outcomes rather than every user interaction. This approach reduces unnecessary data collection while providing meaningful insights into user behaviour, campaign performance, and booking conversions.

By concentrating instrumentation efforts strictly on milestone indicators (such as multi-step validation completions, telephone links, or guide requests), the marketing operations team receives reliable analytics data. This structural approach directly supports reliable conversion signals, allowing accurate evaluation of cost-per-acquisition metrics, quick mapping of drop-off bottlenecks, and clean pipeline validation to drive business strategy.

5. Booking Funnel Tracking Architecture

Tracking each booking step helps the marketing team identify exactly where users abandon the booking process. Below is the behavioral tracking strategy for the OrthoNow appointment booking interface:



Measurement Logic: Drop-off Evaluation Note: Abandonment metrics are quantified analytically by checking volume shifts between successive steps. GTM captures user data at the step-validation boundary, pinpointing structural issues (such as user resistance to completing specific validation fields) quickly.

Funnel Progression Node Specifications

Booking Start (Form Initialization)

Step Name: Appointment Form Opened

GTM Trigger: ce - appointment_booking_start

When Trigger Fires: Fires when a user clicks into or interacts with the first input field in the booking module.

GA4 Event Mapping: Maps directly to GA4 Event: appointment_booking_start

Parameters Collected: form_id, booking_session_id

Step 1 (Clinic & Specialty Selection)

Step Name: Clinic and Specialty Selected

GTM Trigger: ce - appointment_booking_step (Condition: dlvs - step_number equals 1)

When Trigger Fires: Fires exactly when the user selects a clinic location and specialty, then clicks the 'Next' button to proceed to Step 2.

GA4 Event Mapping: Maps directly to GA4 Event: appointment_booking_step

Parameters Collected: step_number, step_name, clinic_location, specialty, form_id, booking_session_id

Step 2 (Patient Details Input)

Step Name: Patient Details Entered

GTM Trigger: ce - appointment_booking_step (Condition: dlvs - step_number equals 2)

When Trigger Fires: Fires exactly when the user enters their Name, Phone, and Preferred Date, then clicks the 'Next' button to view the confirmation screen.

GA4 Event Mapping: Maps directly to GA4 Event: appointment_booking_step

Parameters Collected: step_number, step_name, clinic_location, specialty, form_id, booking_session_id

Step 3 (Booking Confirmation / Success)

Step Name: Booking Confirmed

GTM Trigger: ce - appointment_booking_complete

When Trigger Fires: Fires when the user reviews their details on the final step, clicks 'Confirm Booking', and the database backend API successfully returns an order/appointment ID.

GA4 Event Mapping: Maps directly to GA4 Event: appointment_booking_complete

Parameters Collected: step_number, step_name, clinic_location, specialty, form_id, booking_session_id, appointment_id

6. dataLayer JSON Reference Specifications

The frontend engineering framework must push structured data streams following these formatting layouts to guarantee accurate asynchronous queue evaluations:

Code Payload Matrix: Booking Start Payload

```
{
  "event": "appointment_booking_start",
  "form_id": "appointment_booking",
  "booking_session_id": "orthonow_987654321"
}
```

Field Key	Purpose	Business Reason
event	The custom event name GTM listens for.	Allows triggers to fire instantly upon application lifecycle change.
form_id	Static text identifying the component object.	Differentiates this framework from generic global or footer inputs.
booking_session_id	Connects all booking steps within one user session.	Links multi-step inputs to an isolated user session.

Code Payload Matrix: Booking Step 1 Completion

```
{
  "event": "appointment_booking_step",
  "step_number": 1,
  "step_name": "location_specialty_selected",
  "clinic_location": "Delhi",
  "specialty": "Orthopedics",
  "form_id": "appointment_booking",
  "booking_session_id": "orthonow_987654321"
}
```

Field Key	Purpose	Business Reason
event	The custom event name GTM listens for.	Allows triggers to fire instantly upon application lifecycle change.
step_number	Identifies the booking step.	Provides ordering context for drop-off calculation models.
step_name	Human-readable description of the step.	Ensures team alignment during diagnostic reporting sweeps.
clinic_location	Used to compare clinic performance.	Allows data analysis to slice funnel drops by region.
specialty	Helps analyze demand by medical specialty.	Surfaces operational volume demand trends across disciplines.

Code Payload Matrix: Booking Step 2 Completion

```
{
  "event": "appointment_booking_step",
  "step_number": 2,
  "step_name": "patient_details_entered",
  "clinic_location": "Delhi",
  "specialty": "Orthopedics",
  "form_id": "appointment_booking",
  "booking_session_id": "orthonow_987654321"
}
```

Field Key	Purpose	Business Reason
event	The custom event name GTM listens for.	Allows triggers to fire instantly upon application lifecycle change.
step_number	Identifies the booking step.	Provides ordering context for drop-off calculation models.
step_name	Human-readable description of the step.	Ensures team alignment during diagnostic reporting sweeps.

Code Payload Matrix: Step 3 (Booking Confirmation / Success)

```
{
  "event": "appointment_booking_complete",
  "step_number": 3,
  "step_name": "booking_confirmed",
  "clinic_location": "Delhi",
  "specialty": "Orthopedics",
  "form_id": "appointment_booking",
  "booking_session_id": "orthonow_987654321",
  "appointment_id": "APT-2026-00941"
}
```

Field Key	Purpose	Business Reason
event	The custom event name GTM listens for.	Allows triggers to fire instantly upon application lifecycle change.
appointment_id	Unique transaction ID to eliminate double-counting conversions.	De-duplicates conversions inside GA4 and Google Ads pipeline structures.

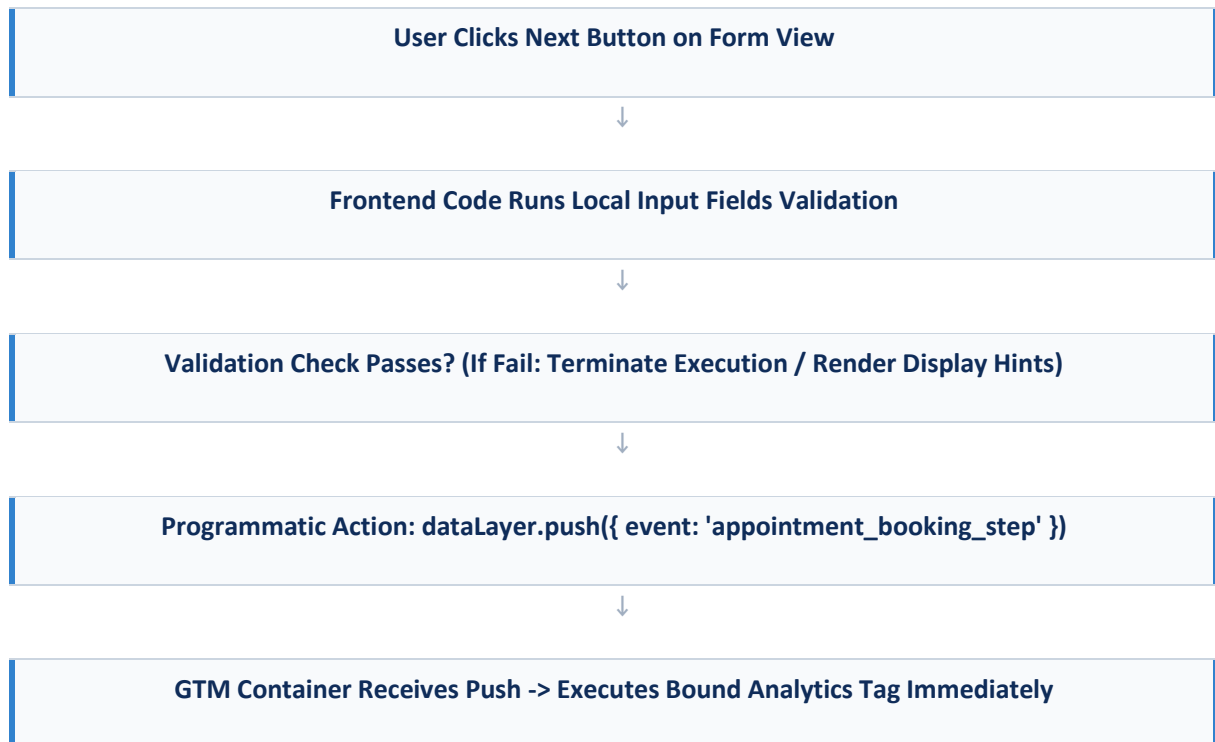
7. GTM Infrastructure & Form Detection Framework

The configuration framework orchestrates data capture from local user interactions, routes it securely into GTM, and distributes clean payloads out to all critical target channels:



Why GTM Cannot Detect Multi-Step Forms Natively

Standard built-in GTM tracking triggers (such as generic form submission listeners) require standard HTML browser submit actions to execute, or visible changes to the website URL route structure to track step progress. Modern multi-step web forms are engineered as dynamic applications using frontend JavaScript frameworks. These structures process state updates locally within browser memory without refreshing pages or generating distinct URL alterations between form views. GTM cannot natively track these internal code steps or differentiate a genuine step progression from an invalid click without a dedicated dataLayer data layout engine acting as an official source of truth.



Operational Roles Definition

- **Frontend Developer:** Responsible for inserting the precise `dataLayer.push()` code blocks into the frontend code structure exactly where step data passes validation rules and updates the display.
- **GTM Developer / Implementation Specialist:** Defines formatting rules, sets variable keys, creates custom event triggers, and maps data streams directly into target metrics paths.

Developer Handoff Brief: Funnel Step 2 Integration

To: Frontend Engineering Team

Context: Tracking Patient Progress in Appointment Booking Funnel

Action Required: Implement a `dataLayer.push()` snippet on the appointment booking module upon the successful completion of Step 2 (Patient Details).

Deployment Timing

The tracking script must execute within the success callback function of the Step 2 view. Run this code only after the user selects the 'Next' control AND frontend validation confirms all inputs (Name, Phone number, and Preferred Date fields) are complete and formatted correctly. This script must

run before any layout logic processes the Step 3 confirmation screen layout. Do not tie this execution to raw clicks on unverified element regions.

Code Payload Matrix: Step 2 Frontend Execution Code

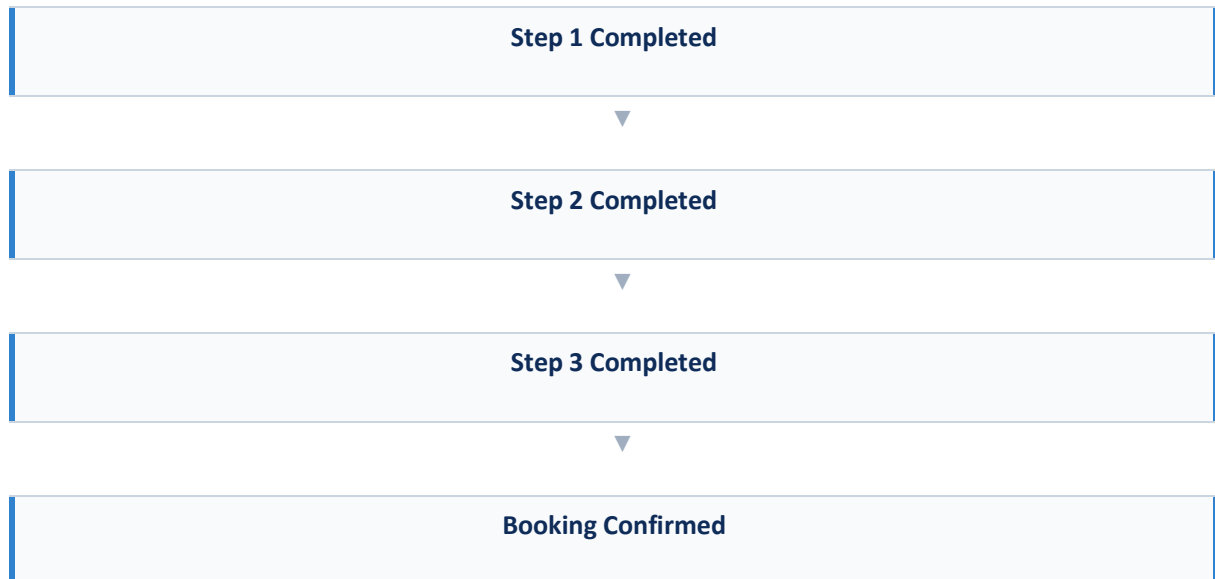
```
window.dataLayer = window.dataLayer || [];  
window.dataLayer.push({  
  'event': 'appointment_booking_step',  
  'step_number': 2,  
  'step_name': 'patient_details_entered',  
  'clinic_location': document.querySelector('#clinic-select').value ||  
  'Unknown', // Persist from Step 1 selection  
  'specialty': document.querySelector('#specialty-select').value || 'Unknown',  
  // Persist from Step 1 selection  'form_id': 'appointment_booking',  
  'booking_session_id': window.bookingSessionId // Match the session ID  
  instantiated at Step 1  
});
```

Technical Validation Guidelines

- Do not push PII (Personally Identifiable Information) under any circumstance. Patient names, actual telephone lines, or explicit clinical calendar date values must never be routed to the dataLayer.
- Verify that dictionary strings use matching case formatting. Use lower_case naming rules consistently across all declared properties.

8. GA4 Funnel Exploration Configuration

To map, isolate, and address completion issues within the appointment flow, configure a Funnel Exploration workspace layout inside GA4 using these parameters:



Analysis Framework: GA4 Funnel Exploration uses these events to visualize user progression through the booking process and identify where users abandon the journey.

Technique Configuration Values

- **Technique:** Funnel Exploration Workspace Grid.
- **Funnel Type:** Standard Linear Flow Charting.
- **Open vs. Closed Funnel:** Closed Funnel configuration. This rule forces metrics calculation models to count a user only if they enter the tracking footprint at Step 1. Users matching mid-funnel markers out of order do not skew calculation rates or distort pipeline tracking records.
- **Visualization and Breakdown:** Standard Bar Chart Funnel. Enable the 'Show Elapsed Time' configuration toggle to monitor the average duration required by users to move between step validation blocks.

9. Google Ads Conversion Ingestion Strategy

Conversion Setup: Primary Account Conversion Target: appointment_booking_complete

Strategic Justification

The appointment_booking_complete event indicates a definitive, high-value user action: a confirmed clinical slot reservation. This interaction represents successful acquisition. Importing this specific event as a Primary Conversion tells Google Ads optimization models to focus budget on users who pass medical validation, rather than general site visitors.

Why Mid-Funnel Actions Remain Secondary

Choosing `appointment_booking_complete` instead of `booking_started` or `booking_step` avoids training the bidding model on partial intents. Setting actions like `appointment_booking_start` or individual booking steps as primary conversions splits optimization focus. This can cause automated bidding systems to maximize top-of-funnel clicks that drop off early due to form friction, rather than focusing on completed actions.

While actions like `click_to_call` or guide requests demonstrate user intent, they do not guarantee clinical revenue or actual capacity optimization. Keeping them as secondary actions allows the marketing team to evaluate full channel attribution paths without degrading automation signals.

Google Ads Smart Bidding Integration

By routing a single conversion goal, Maximize Conversions and Target CPA automated bidding scripts receive clear intent data. Machine learning loops map traits (such as search intent modifiers, regional device profiles, and interaction hours) directly to successful completions, improving patient acquisition cost-effectively.

10. Technical Deployment Assumptions

- The appointment form functions as an asynchronous client-side component, processing view updates without causing full page updates or generating distinct URL changes.
- Engineering channels can access frontend source files and bind tracking events directly within code success callback blocks.
- The active Google Tag Manager tracking container script is integrated into the global header files across all target domains.
- The base Google Tag configuration is active, establishing proper tracking data routing rules before any custom data configurations execute.

11. Architectural Conclusion

This GTM event schema provides a scalable tracking framework for the OrthoNow website. By combining structured `dataLayer` events, Google Tag Manager, and Google Analytics 4, the marketing team can accurately measure user behaviour, identify booking funnel drop-offs, optimize advertising campaigns, and improve patient acquisition. The proposed design also supports future integrations with CRM platforms such as HubSpot without requiring major changes to the tracking architecture.